On the Hardy-Littlewood maximal function in Sobolev spaces

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Preliminaries

Vitali's cobering lemma

Let $E \subseteq \mathbb{R}^n$ be a measurable set, such that $E \subseteq \bigcup_\alpha B_{r_\alpha}(x_\alpha)$ with the family of open balls $\{B_{r_\alpha}(x_\alpha)\}_\alpha$ that satisfies $\sup_\alpha r_\alpha = c_0 < \infty$.
Then there exists a subfamily $\{B_{r_j}(x_j)\}_{j\in\mathbb{Z}^+}$ of disjoint and numerable elements such that

$$|E| \leq 5^n \sum_{j=1}^{\infty} |B_{r_j}(x_j)|.$$

Distribution function

Let (X,μ) be a measurable space, and $f : X \to \mathbb{C}$ be a measurable function.
We set the distribution function of f associated with μ the function as follows

$$a_f : [0,\infty] \to [0,\infty]$$
$$\lambda \to \mu(\{x \in X : |f(x)| > \lambda\}) = \mu(E_f^\lambda).$$

Weak Spaces L^p

For $1 \leq p \leq \infty$, we denote by $L^{p*}(X,\mathcal{A},\mu)$ the space of all measurable functions $f : X \to \mathbb{C}$ such that

$$\|f\|^*_{L^p(X)} = \sup_{\lambda>0} \lambda (a_f(\lambda))^{1/p} < \infty.$$

Equivalently, $L^{p*}(X,\mathcal{A},\mu)$ consists of all functions f that satisfy

$$a_f(\lambda) \leq \frac{C}{\lambda^p}.$$

Some types of inequalities

Let (X,μ) and (Y,ν) be two measurable spaces, and let T be an operator from $L^p(X,\mu)$ to the space of measurable functions on Y with values in $\mathbb{C}$, that is $T : L^p(X,\mu) \to \mathcal{M}(Y,\mathbb{C})$.

i. We say that T is (p,q)-weak (with $q < \infty$) if for every $\lambda > 0$, there exists $C > 0$ such that $a_{Tf}(\lambda) \leq \left(\frac{C\|f\|_{L^p(X)}}{\lambda}\right)^q$.

iii. We say that T is (p,q)-strong if the operator is bounded from $L^p(X,\mu)$ to $L^q(Y,\nu)$.

The Marcinkiewicz Interpolation Theorem (diagonal Case)

Let (X,μ) and (Y,ν) be measure spaces, $1 \leq p_0 < p_1 \leq \infty$, and let T be a sublinear operator from $L^{p_0}(X,\mu) + L^{p_1}(X,\mu)$ to measurable functions on Y that is weak (p_0,p_0) and is weak (p_1,p_1). Then T is strong (p,p) for $p_0 < p < p_1$.

The Sobolev space

Given $1 \leq p \leq \infty$, we define $W^{k,p}(U)$ by all functions $u : U \to \mathbb{R}$ such that for each multiindex α with $|\alpha| \leq k$, $\partial^\alpha u$ exist in the weak sense and belongs to $L^p(U)$.
If $u \in W^{k,p}(U)$, we define the norm

$$\|f\|_{W^{1,p}(U)} = \|f\|_{L^p(U)} + \|\nabla f\|_{L^p(U)}.$$

The Hardy-Littlewood Maximal Function

Given $f \in L^1_{loc}(\mathbb{R}^n)$, we define $\mathcal{M}f(x)$, the Hardy-Littlewood maximal function of a f by

$$\mathcal{M}f(x) = \sup_{r>0} \frac{1}{|B_r(x)|}\int_{B_r(x)} |f(y)|\,dy = \sup_{r>0}\left(|f| * \frac{1}{|B_r(0)|}\chi_{B_r(0)}\right)(x).$$

Results

Affirmation. $\mathcal{M}$ is an operator (∞,∞)-strong.
Theorem. The operator $\mathcal{M}$ are $(1,1)$-week.
As consecuence of the Marcinkiewicz interpolation theorem and the previous results, we affirm that $\mathcal{M}$ is an operator (p,p)-strong for all $1 < p \leq \infty$.
Note that given $f \in L^1$, $\mathcal{M}f \in L^1$ if and only if $f = 0$. Therefore, does not hold $(1,1)$-strong.

Lebesgue differentiation theorem. *Let be $f \in L^1_{loc}(\mathbb{R}^n)$, then*

$$\lim_{r\to 0^+} \frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)\,dy = f(x) \quad a.e.$$

Theorem. *Let ϕ be a radial, positive and nonincreasing function of $r = |x| \in [0,\infty)$, then taking $\phi_t(x) = t^{-n}\phi(t^{-1}x)$ it is satisfies that $\sup_t |\phi_t * f(x)| \leq \|\phi\|_1 \mathcal{M}f(x)$.*
Suppose ϕ be a simple functions, then

$$\begin{aligned}|\phi * f(x)| &= \left|\left(\sum_{j=1}^{\infty} c_j \chi_{B_{r_j}(x_j)} * f\right)(x)\right| \\ &= \left|\sum_{j=1}^{\infty} c_j |B_{r_j}(x_j)| \left(\frac{1}{|B_{r_j}(x_j)|}\chi_{B_{r_j}(x_j)} * f\right)(x)\right| \\ &\leq \|\phi\|_1 \mathcal{M}f(x).\end{aligned}$$

Since ϕ_t is a dilation of ϕ, the same procedure can be followed and the same result reached. As a consecuence, we have that maximal function $\sup_t |\phi_t * f(x)|$ are $(1,1)$-week and (p,p)-strong with $1 < p \leq \infty$. This result allows us to generalize (without using sequences $\{t_n\}$) the convergence results for approximations of the identity.

Examples

- For $t > 0$, the heat kernel in $\mathbb{R}^n$ defined by $K_t(x) = \frac{1}{(4\pi t)^{\frac{n}{2}}} e^{-\frac{|x|^2}{4t}}$ satisfies the convergence result mentioned above.
- For $t > 0$, the Poisson kernel in $\mathbb{R}^n$ is defined by $P_t(x) = c_n \frac{t}{(t^2+|x|^2)^{\frac{n+1}{2}}}$, where $c_n > 0$ is the normalization constant such that $\|P_t\|_{L^1(\mathbb{R}^n)} = 1$ satisfies the convergence results.

Theorem [5]. *It holds that*

$$\mathcal{M} : W^{1,p}(\mathbb{R}^n) \to W^{1,p}(\mathbb{R}^n)$$

defines a bounded operator when $1 < p \leq \infty$.
(Sketch of the proof) It is enough to show that if $u \in W^{1,p}(\mathbb{R}^n)$ with $1 < p < \infty$, then $\mathcal{M}f(x)$ has a weak derivative and satisfies

$$|\partial_{x_i}(\mathcal{M}u)| \leq \mathcal{M}\partial_{x_i}u.$$

On the other hand, when $p = \infty$, we will show that $\mathcal{M}f$ is Lipschitz. To see the first inequality, it is enough to observe that if $u \in W^{1,p}(\mathbb{R}^n)$, then $|u| \in W^{1,p}(\mathbb{R}^n)$ and that

$$\partial_{x_i}(|u| * \chi_r) = \partial_{x_i}|u| * \chi_r.$$

with $\chi_r(x) = \frac{\chi_{B_r(0)}}{|B_r(0)|}(x)$, for every $i = 1,\cdots,n$.
Take a sequence $r_j > 0$ with $j = 1,2,\cdots$ as an enumeration of the positive rationals and define the functions $v_k : \mathbb{R}^n \to \mathbb{R}$, for $k = 1,2\cdots$, by

$$v_k(x) = \max_{1\leq j\leq k}\left(|u| * \chi_{r_j}\right)(x).$$

Note that the sequence of functions $\{v_k\} \subset W^{1,p}(\mathbb{R}^n)$ is increasing and bounded; moreover, by construction we have pointwise convergence to $\mathcal{M}\partial_{x_i}u$ since

$$\sup_k |\partial_{x_i}v_k| \leq \sup_k \max_{1\leq j\leq k} |\partial_{x_i}|u| * \chi_{r_j}| \leq \mathcal{M}\partial_{x_i}u.$$

Thus, taking the $L^p(\mathbb{R}^n)$ norm, we obtain

$$\|\partial v_k\|_p \leq \sum_{i=1}^n \|\partial_{x_i}v_k\|_p \leq \sum_{i=1}^n \|\mathcal{M}\partial_{x_i}u\|_p.$$

Then, using the Hardy–Littlewood theorem, it follows that

$$\|v_k\|_{1,p} \leq \|\mathcal{M}u\|_p + \sum_{i=1}^n \|\mathcal{M}\partial_{x_i}u\|_p \leq c_p\|u\|_p + c_p\sum_{i=1}^n \|\partial_{x_i}u\|_p < \infty,$$

hence, for every $k = 1,2,\cdots$, the sequence $\{v_k\} \subset W^{1,p}(\mathbb{R}^n)$ is an increasing and bounded sequence in $W^{1,p}(\mathbb{R}^n)$. Therefore, since the space $W^{1,p}(\mathbb{R}^n)$ is reflexive, there exists a weakly convergent subsequence $\{v_{k_i}\}$; however, since the sequence $\{v_k\}$ is increasing, the whole sequence converges weakly to $\mathcal{M}|u|$, and by the same argument $\{\partial_{x_i}v_k\}$ converges weakly to $\mathcal{M}\partial_{x_i}|u|$. This allows us to conclude that

$$|\partial_{x_i}\mathcal{M}u| \leq \mathcal{M}\partial_{x_i}u.$$

Let $f \in W^{1,\infty}(\mathbb{R}^n)$. We shall show that $\mathcal{M}(f)$ is Lipschitz continuous. For this, let $x,y \in \mathbb{R}^n$ and $r > 0$. By making a change of variables to center the balls at 0 and using some properties of the absolute value, we have

$$\frac{1}{|B_r(x)|}\int_{B_r(x)} |f(z)|\,dz - \mathcal{M}f(y) \leq \frac{1}{|B_r(0)|}\int_{B_r(0)} \Big||f(x+w) - f(y+w)|\Big|\,dw.$$

Since $f \in W^{1,\infty}(\mathbb{R}^n)$, by the mean value theorem it follows that

$$\big|f(x+w) - f(y+w)\big| \leq \|\nabla f\|_{L^\infty}|x-y|.$$

We conclude that

$$\frac{1}{|B(x,r)|}\int_{B(x,r)} |f(z)|\,dz - \mathcal{M}f(y) \leq \|\nabla f\|_{L^\infty}|x-y|.$$

Taking the supremum over $r > 0$, we obtain from the previous inequality

$$\mathcal{M}f(x) - \mathcal{M}f(y) \leq \|\nabla f\|_{L^\infty}|x-y|.$$

And by swapping the roles of x and y, we arrive at

$$\big|\mathcal{M}f(y) - \mathcal{M}f(x)\big| \leq \|\nabla f\|_{L^\infty}|x-y|.$$

From the above, it follows that $\mathcal{M}f$ is Lipschitz continuous. Therefore, the characterization of $W^{1,\infty}(\mathbb{R}^n)$ shows that $\mathcal{M}f \in W^{1,\infty}(\mathbb{R}^n)$ and

$$\|\nabla\mathcal{M}f\|_{L^\infty} \leq \|\nabla f\|_{L^\infty}.$$

Future Work

- Study additional properties of the Hardy–Littlewood maximal function in Sobolev spaces. For example, consult the results presented in [7], pages 25-67.
- Analyze other types of maximal functions. For example, consider the heat kernel

  $$K_t(x) = \frac{1}{(4\pi t)^{\frac{d}{2}}} e^{-\frac{|x|^2}{4t}}.$$

  Study the associated maximal operator,

  $$\mathcal{K}f(x) := \sup_{t>0}\big(f * K_t\big)(x),$$

  in Sobolev spaces $W^{1,p}(\mathbb{R}^n)$. For this, one may consult the articles [2], [1].

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